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Nortwind Interactive Sales

Dashboard POWER BI SALES ANALYSIS

ACTIVITY

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POWER BI SALES ANALYSIS ACTIVITY

1. INTRODUCTION & OBJECTIVES

For this activity, I created an interactive Sales Analysis Dashboard using Microsoft Power BI. The purpose of the dashboard is to convert sales data into meaningful visual information and provide an easy way to analyse business performance. The report presents important sales measures through KPI cards, charts, filters, a tooltip page, and a drill-through page.

The main objective is to analyse overall sales performance, order activity, quantity sold, freight values, customer information, employee-wise orders, and product category performance. The report also provides detailed analysis by using interactive filtering and drill-through functionality.

2. DATASET & DATA MODEL

Tables Used in the Analysis

The dashboard is created using multiple related tables containing information about customers, employees, products, categories, orders, and other sales-related details.

The major information used in the report includes:

- Customer information
- Employee information
- Product information
- Product category information
- Order information
- Order detail information
- Freight information

Table Relationships

The tables were connected using appropriate relationships in Power BI. These relationships allow information from different tables to work together when creating visuals and calculations.

For example, order information can be analysed together with employee, product, category, and customer information.

Data Preparation

Before creating the visuals, the required fields were checked and the tables were properly connected. The data was then used to create the required measures and visualizations for the dashboard.

3. KEY PERFORMANCE INDICATORS



Total Sales

The **Total Sales** KPI represents the overall sales value available in the report. The dashboard displays approximately **1.27M** in total sales.

Total Orders

The **Total Orders** KPI shows the total number of orders included in the analysis.

Total Orders = 830

Total Quantity Sold

The **Total Quantity Sold** KPI represents the total quantity of products sold.

Total Quantity Sold = 51K

Total Freight

The **Total Freight** KPI represents the total shipping or freight amount.

Total Freight = 64.94K

Average Freight

The **Average Freight** KPI shows the average freight value.

Average Freight = 78.24

Number of Customers

The dashboard also displays the total number of customers included in the analysis.

Number of Customers = 91

4. ORDER TREND ANALYSIS

Total Orders by Month

A line chart is used to display the **Total Orders by Month**. The chart presents order activity from January to December and shows how the order volume changes throughout the year.

Monthly Order Pattern

The visualization shows higher order activity around **March and April**, followed by a noticeable decline during **May and June**. After this decline, order activity increases again towards the later months.

This visual provides a clear understanding of the changes in order activity during different months.

5. EMPLOYEE ORDER ANALYSIS

Total Orders by Employee

A horizontal bar chart is used to compare the Total Orders handled by employees.

The visual includes employees such as Margaret Peacock, Janet Leverling, Nancy Davolio, Laura Callahan, Andrew Fuller, Robert King, Michael Suyama, and Anne Dodsworth.

Employee-Wise Comparison

The chart allows the order volume of each employee to be compared. From the displayed visualization, Margaret Peacock has one of the highest order counts.

This analysis helps show how the order workload is distributed among employees.

6. PRODUCT CATEGORY ANALYSIS

Sales by Product Category

A column chart is used to display **Sales by Product Category**. The visual compares the sales contribution of the different product categories.

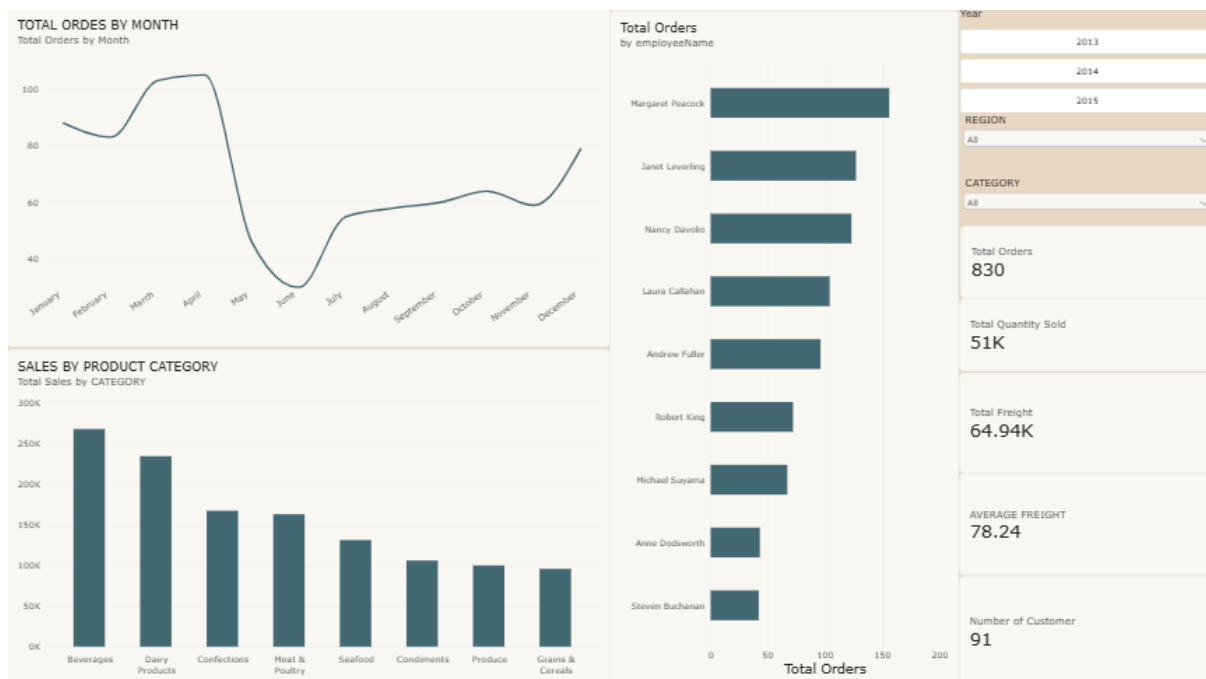
PRODUCT SALES SUMMARY			
productName	CATEGORY	Total Sales	Total Quantity Sold
Alice Mutton	Meat & Poultry	32,698.38	978
Aniseed Syrup	Condiments	3,044.00	328
Boston Crab Meat	Seafood	17,910.63	1103
Camembert Pierrot	Dairy Products	46,825.48	1577
Carnarvon Tigers	Seafood	29,171.88	539
Chai	Beverages	12,788.10	828
Chang	Beverages	16,355.96	1057
Chartreuse verte	Beverages	12,294.54	793
Chef Anton's Cajun Seasoning	Condiments	8,567.90	453
Chef Anton's Gumbo Mix	Condiments	5,347.20	298
Chocolade	Confections	1,368.71	138
CÔte de Blaye	Beverages	1,41,396.74	623
Escargots de Bourgogne	Seafood	5,881.68	534
Filo Mix	Grains & Cereals	3,232.95	500
Flotemysost	Dairy Products	19,551.03	1057
Geltost	Dairy Products	1,648.13	755
Genen Shouyu	Condiments	1,784.83	122
Gnocchi di nonna Alice	Grains & Cereals	42,593.06	1263
Gorgonzola Telino	Dairy Products	14,920.88	1397
Grandma's Boysenberry Spread	Condiments	7,137.00	301
Gravad lax	Seafood	2,688.40	125
Guarana Fantastica	Beverages	4,504.37	1125
Gudbrandsdalsost	Dairy Products	21,942.36	714
Gula Malacca	Condiments	9,915.95	601
GumbOr GummibOrchen	Confections	19,849.14	753
Gustaf's KnackebOrd	Grains & Cereals	7,122.36	348
Ikura	Seafood	20,867.34	742
Inlagd Sill	Seafood	13,458.46	805
Ipoh Coffee	Beverages	23,526.70	580
Jack's New England Clam Chowder	Seafood	8,680.35	981
Konbu	Seafood	4,960.44	891
LakkalikOri	Beverages	15,760.44	981
Laughing Lumberjack Lager	Beverages	2,396.80	184
Total		12,65,793.04	51317

Category-Wise Comparison

The chart shows that sales are not equally distributed among all categories. Some categories have higher sales values, while others contribute comparatively less.

This makes it easier to identify differences in category performance.

7. INTERACTIVE DASHBOARD FILTERS



Year Selection

The dashboard contains a **Year slicer** that allows the user to select different years shown in the report, including **2013, 2014, and 2015**.

Category Selection

A **Category slicer** is also provided to select a particular product category.

Interactive Filtering

When the user changes the slicer selection, the relevant dashboard information changes according to the selected year or category. This makes the report interactive and allows users to focus on specific data.

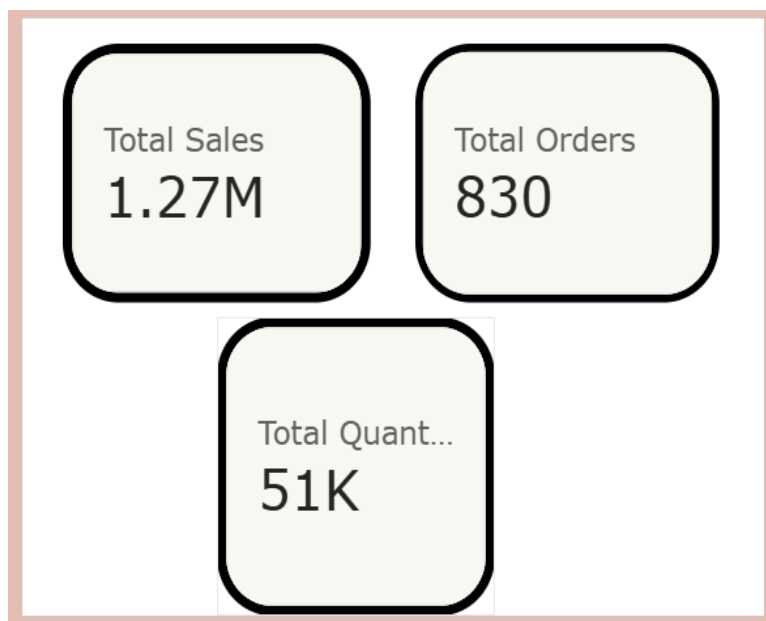
8. CATEGORY TOOLTIP ANALYSIS

Category Tool Tip

A separate **Category Tool Tip** page has been created to display additional information related to the selected category.

The tooltip contains important measures such as:

- Total Sales
- Total Orders
- Total Quantity Sold



Tooltip-Based Information

The tooltip allows the user to obtain additional information while interacting with the dashboard without placing all the information directly on the main report page.

This makes the main dashboard less crowded while still providing additional analytical information.

9. CATEGORY DRILL-THROUGH ANALYSIS

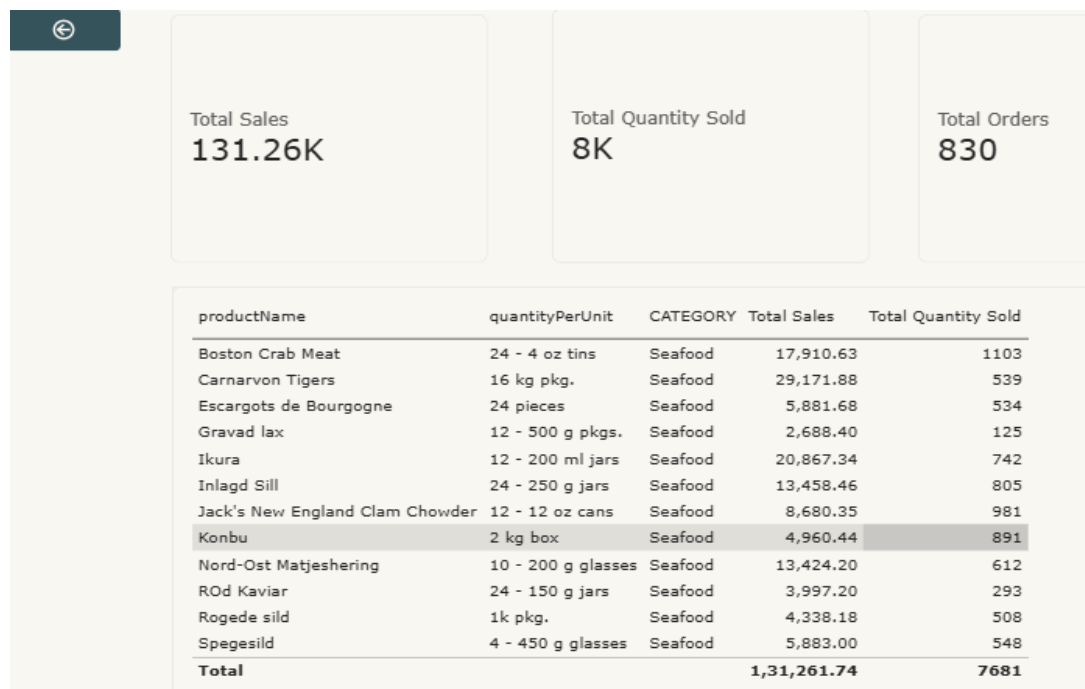
Category Details

A **Category Details** page has been created using the Drill-Through feature. This page provides detailed information about the selected category.

Seafood Performance

For the selected Seafood category, the report displays:

- **Total Sales – 131.26K**
- **Total Quantity Sold – approximately 8K**
- **Total Orders – 830**



productName	quantityPerUnit	CATEGORY	Total Sales	Total Quantity Sold
Boston Crab Meat	24 - 4 oz tins	Seafood	17,910.63	1103
Carnarvon Tigers	16 kg pkg.	Seafood	29,171.88	539
Escargots de Bourgogne	24 pieces	Seafood	5,881.68	534
Gravad lax	12 - 500 g pkgs.	Seafood	2,688.40	125
Ikura	12 - 200 ml jars	Seafood	20,867.34	742
Inlagd Sill	24 - 250 g jars	Seafood	13,458.46	805
Jack's New England Clam Chowder	12 - 12 oz cans	Seafood	8,680.35	981
Konbu	2 kg box	Seafood	4,960.44	891
Nord-Ost Matjeshering	10 - 200 g glasses	Seafood	13,424.20	612
ROd Kaviar	24 - 150 g jars	Seafood	3,997.20	293
Rogede sild	1k pkg.	Seafood	4,338.18	508
Spegesild	4 - 450 g glasses	Seafood	5,883.00	548
Total			1,31,261.74	7681

These KPIs provide a summary of the selected category.

Product Details

The Category Details page contains a table showing product-level information such as:

- Product Name
- Quantity Per Unit
- Category
- Total Sales
- Total Quantity Sold

The table provides detailed information about the products belonging to the selected Seafood category.

10. DASHBOARD INSIGHTS

Overall Sales Performance

The dashboard shows **1.27M in Total Sales**, indicating the overall sales value represented in the report.

Order and Quantity Performance

The report contains **830 Total Orders** and approximately **51K Total Quantity Sold**, providing an overview of both transaction volume and product volume.

Freight Performance

The dashboard shows **64.94K Total Freight** and an **Average Freight of 78.24**.

Customer Overview

The report contains **91 customers**, providing a summary of the customer base represented in the dashboard.

Category and Product Insights

The category visual provides a comparison of sales performance across categories, while the drill-through page allows detailed analysis of the products within the selected category.

11. BENEFITS OF THE VISUALS AND FEATURES USED

- Cards provide a quick view of important values such as total sales, orders, quantity sold, freight, and customers.
- The line chart helps identify monthly order trends and changes over time.
- The bar chart makes it easy to compare the number of orders handled by employees.
- The column chart helps compare sales performance across different product categories.
- Slicers allow users to filter the dashboard based on year and category.
- The tooltip provides additional information without making the main dashboard crowded.

- Drill-through allows users to move from the main dashboard to det

12. LEARNING OUTCOMES

Through this activity, I learned how to create an interactive Power BI dashboard using related business data. I gained practical experience in creating KPI cards, line charts, bar charts, category analysis, slicers, tooltips, and drill-through pages.

I also learned how to move from an overall business summary to detailed category and product-level information. This activity improved my understanding of how Power BI can be used to organize, visualize, and analyse business data effectively.

13. CONCLUSION

The Power BI Sales Analysis Dashboard provides an interactive way to understand sales and order performance. The dashboard combines KPI cards, monthly order analysis, employee-wise order analysis, category-wise sales analysis, slicers, tooltips, and drill-through into a single report.

The main dashboard provides an overall summary, while the Category Details drill-through page allows deeper analysis of the selected Seafood category and its individual products. The tooltip and slicers further improve the interactive nature of the report.